

Addressing Endogeneity in International Marketing Applications of Partial Least Squares Structural Equation Modeling

G. Tomas M. Hult, Joseph F. Hair Jr., Dorian Proksch, Marko Sarstedt, Andreas Pinkwart, and Christian M. Ringle

ABSTRACT

Partial least squares structural equation modeling (PLS-SEM) has become a key method in international marketing research. Users of PLS-SEM have, however, largely overlooked the issue of endogeneity, which has become an integral component of regression analysis applications. This lack of attention is surprising because the PLS-SEM method is grounded in regression analysis, for which numerous approaches for handling endogeneity have been proposed. To identify and treat endogeneity, and create awareness of how to deal with this issue, this study introduces a systematic procedure that translates control variables, instrumental variables, and Gaussian copulas into a PLS-SEM framework. We illustrate the procedure's efficacy by means of empirical data and offer recommendations to guide international marketing researchers on how to effectively address endogeneity concerns in their PLS-SEM analyses.

Keywords: control variable, endogeneity, Gaussian copula, instrumental variable, partial least squares structural equation modeling (PLS-SEM)

The ambition to make causal claims is fundamental to international marketing research. Consider, for example, a sampling of recent research questions addressed in articles published in *Journal of International Marketing*: Is coopetition between foreign and local partners good or bad for international joint venture

performance (Shu, Jin, and Zhou 2017)? How does strategic fit between host–home country similarity and exploration/exploitation strategies influence firm performance (Cui, Walsh, and Zou 2014)? Do macroenvironmental factors such as cultural dimensions and globalization forces moderate the relationship between product diversification and market value of large international firms (Qiu et al. 2015; Tianjiao 2014)?

Analyzing such research questions requires a research design that randomly assigns corporations or customers to different experimental conditions and then tests for differences among the groups (Antonakis et al. 2010). However, as in many other fields of scientific inquiry

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(e.g., Clougherty, Duso, and Muck 2016; Sande and Ghosh 2018), researchers in international marketing seldom undertake randomized controlled experiments. This is not surprising, given that such experiments are prohibitively expensive or are simply impossible to conduct because of managers and firms' unwillingness to be randomly assigned to the treatment and control groups (Bascle 2008). As a consequence, researchers typically rely on survey-based primary data or secondary data (e.g., Nakata and Huang 2005) in which independent variables cannot be exogenously manipulated (e.g., Clougherty, Duso, and Muck 2016).

At the same time, in international marketing, opportunities for endogeneity to be present are widespread throughout empirical research designs and data contexts. As Deng (2016, p. 1) notes, "endogeneity is pervasive and sometimes inevitable in international business and marketing studies." Numerous researchers have, therefore, called for routine consideration of endogeneity in these and related fields (e.g., Hult et al. 2008; Sande and Ghosh 2018).

Endogeneity arises, for example, when a firm's international marketing strategies (e.g., channel selection, entry modes, and advertising spending) are nonrandom and influenced by firm- and industry-level factors. These factors may include a firm's corporate culture and organizational capabilities or market forces such as industry growth and competition intensity (Jin, Zhou, and Wang 2016). These factors are typically difficult to measure and include in statistical models (e.g., Hult et al. 2008). Similarly, a multitude of factors, which are difficult to consider in full, potentially influence a firm's international marketing performance (i.e., the outcome of a firm's decision making regarding internationalization activities; Souchon et al. 2016). In these situations, if an omitted construct or variable affects both the dependent variable and one or more independent variable(s) in the regression model, the condition will induce a correlation between the independent variable(s) and the error term (Wooldridge 2013). Thus, variation in the omitted construct or variable will manifest in the error term, thereby violating a key causal modeling assumption of regression analysis: the error term is assumed to have an expected value of zero (Wooldridge 2013). As a consequence, the coefficient estimates from standard regression analyses are biased and inconsistent, thereby becoming causally uninterpretable and potentially triggering Type I and Type II errors (Papies, Ebbes, and van Heerde 2017).

Dealing with endogeneity has been extensively discussed in the marketing literature, especially with respect to different forms of regression and panel models (e.g., Ebbes, Papies,

and van Heerde 2011; Park and Gupta 2012; Rossi 2014) as well as conjoint analysis (e.g., Liu, Otter, and Allenby 2007). But little research addresses endogeneity in structural equation modeling (SEM). Although several studies have discussed endogeneity in the context of factor-based SEM (e.g., Bollen, Kolenikov, and Bauldry 2014; Kirby and Bollen 2009), there is a paucity of research on this topic in partial least squares structural equation modeling (PLS-SEM; Wold 1982), the use of which has recently gained momentum in international marketing (Richter et al. 2016) and business research in general (e.g., Ali et al. 2018; Nitzl 2016; Ringle et al. 2018). Some researchers even claim that PLS-SEM does not allow endogeneity to be addressed at all (e.g., Antonakis et al. 2010; McIntosh, Edwards, and Antonakis 2014; Rönkkö and Evermann 2013). This assertion is astonishing and inaccurate given that PLS-SEM is grounded in regression analysis (Lohmöller 1989, Chapter 2), for which numerous approaches for handling endogeneity exist (e.g., Ebbes et al. 2005; Park and Gupta 2012; Staiger and Stock 1997). Indeed, Benitez, Henseler, and Roldán (2016) recently made an advance in this direction by combining the standard PLS-SEM algorithm with the two-stage least squares (2SLS) method, but they did not consider variables that control for endogeneity's sources.

Overall, researchers have not yet developed a comprehensive framework that considers the multitude of approaches applicable in PLS-SEM and offers guidelines for assessing endogeneity in this context. It is therefore not surprising that PLS-SEM applications do not use approaches for assessing endogeneity. Specifically, our review of articles published in the ten years between 2008 and 2017 in *Journal of International Marketing*, *International Marketing Review*, and *Journal of International Business Studies*—the three highest-ranked international marketing journals in Hult, Reimann, and Schilke's (2009) journal ranking—shows that none of the 43 reviewed PLS-SEM studies tested for endogeneity (Appendix).

Addressing this gap in research, this study explains and illustrates how to deal with endogeneity in PLS-SEM-based international marketing studies. We discuss endogeneity's roots and consequences, focusing on omitted variables as the principal, and most commonly examined, source of endogeneity (Bascle 2008; Clougherty, Duso, and Muck 2016).¹ We then describe the different methods of (statistically) responding to endogeneity in regression models and translate these to a PLS-SEM context. Drawing on the previously mentioned discussion, we offer a systematic

¹Other sources of endogeneity include measurement error and nonrecursive relationships between constructs (Wooldridge 2010).

procedure for identifying and treating endogeneity in PLS-SEM and illustrate its use in an empirical example of corporate reputation, a concept frequently examined in international marketing and business research.

TREATING ENDOGENEITY IN REGRESSION MODELS

Although endogeneity can have various roots, such as measurement errors, simultaneous causality, common method variance, and (un)observed heterogeneity (Bascle 2008; McIntosh, Edwards, and Antonakis 2014; Papies, Ebbes, and van Heerde 2017; Sande and Ghosh 2018), endogeneity problems most often arise from the omission of variables that correlate with one or more independent variable(s) and the dependent variable(s) in the regression model (e.g., Bijmolt, van Heerde, and Pieters 2005; Rossi 2014). Omitting such variables induces a correlation between the corresponding independent variables and the dependent variables' error term (Wooldridge 2010). That is, the independent variables then explain not only the dependent variable but also the error in the model.

Consider the following regression model, where y represents the dependent variable, x_1 and x_2 are independent variables, β_0 is the intercept, β_1 and β_2 are the regression coefficients of x_1 and x_2 , and ε is the error term:

$$(1) \quad y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon.$$

Let us assume that the independent variable x_2 is uncorrelated with ε (i.e., x_2 is exogenous), whereas x_1 is endogenous because it is correlated with the error term ε (i.e., $\text{Cov}(x_1, \varepsilon) \neq 0$). Thus, endogeneity occurs in this regression model example. Note that this terminology, which is used in the endogeneity literature (Wooldridge 2010), should not be confused with the terms used for the explanation of PLS path models with exogenous (i.e., independent) and endogenous (i.e., dependent) constructs (Chin 1998). In the following, and in line with the literature, we use *endogenous* and *exogenous* to identify variables that endogeneity does or does not affect; we use *dependent* and *independent* to identify constructs that explain other constructs in the PLS path model or are explained by them.

A straightforward way of handling, or at least reducing, endogeneity is to specify a set of control variables (Bernierth and Aguinis 2016) accounting for a part of the dependent variable's variance (Ebbes, Papies, and van Heerde 2017). Although carefully chosen control variables often alleviate endogeneity problems substantially (Papies, Ebbes, and van Heerde 2017), the success of this endeavor depends on the researcher's ability to identify relevant control variables and

collect corresponding data. Furthermore, specifying a great number of control variables does not necessarily account for all endogeneity in the model (Bascle 2008; Reeb, Sakakibara, and Mahmood 2012) and contradicts the basic philosophy of Occam's razor (i.e., the principle of parsimony) on which the scientific discipline rests.

Despite careful selection of control variables, researchers need to apply a statistical approach to address endogeneity if endogeneity could be a problem. Two broad statistical approaches have been developed to examine the presence of endogeneity: the instrumental variable (IV) approach and IV-free approaches (Papies, Ebbes, and van Heerde 2017).

Instrumental Variable Approach

In econometrics and other areas of applied research, the IV approach is the technique of choice when dealing with endogeneity (e.g., Bound, Jaeger, and Baker 1995; Staiger and Stock 1997; Villas-Boas and Winer 1999). This approach uses information in additionally specified IVs to decompose the endogenous independent variables' variance into two parts (Bascle 2008): the exogenous part, which does not correlate with the error term, and the endogenous part, which correlates with the error term. The estimation of the regression model then uses only the exogenous part, which endogeneity does not affect, to provide unbiased and consistent regression estimates. The IV approach is usually implemented by means of the 2SLS algorithm. This algorithm's first stage involves regressing an independent variable on an IV and computing the predicted values from this equation. These values are then used as input to reestimate the original model in the second stage.

To illustrate the approach, consider the regression model example given in Equation 1. Let us assume that the independent variable x_1 is endogenous, whereas x_2 is exogenous. The first stage of the IV approach involves regressing the endogenous variable x_1 on the exogenous variable x_2 and the IV z :

$$(2) \quad x_1 = \gamma_0 + \gamma_1 z + \gamma_2 x_2 + \zeta.$$

Here, γ_0 represents the intercept, γ_1 the regression coefficient of z , γ_2 the regression coefficient of x_2 , and ζ the error term. Estimating Equation 2 yields predicted values of x_1 (i.e., $\hat{x}_1$), which express the exogenous part of the independent variable x_1 . In the second stage, we replace x_1 in the original regression model (Equation 1) with $\hat{x}_1$ to estimate the following:

$$(3) \quad y = \beta'_0 + \beta'_1 \hat{x}_1 + \beta'_2 x_2 + \varepsilon'.$$

Endogeneity occurs when there is a significant difference between β_1 (Equation 1) and β'_1 (Equation 3), with the latter

representing the influence of the independent variable's exogenous part on y . To test for this difference, researchers use the Hausman test (Hausman 1978). If the test indicates a significant difference between the β estimates, researchers should revert to the endogeneity-corrected estimates (Equation 3). If the Hausman test does not indicate a significant degree of endogeneity, researchers should use the noncorrected estimates from the original regression model (Equation 1).

To work effectively, the IVs must meet two criteria: the instrument relevance criterion and the exclusion restriction. First, according to the instrument relevance criterion, the correlation of each IV with the endogenous variable should be strong, which means that the IV's effect γ_1 on the dependent variable x_1 in Equation 2 must be pronounced and significant (Papies, Ebbes, and van Heerde 2017). If it is not, the corrected β coefficient estimates (Equation 3) may be considerably biased (Rossi 2014). But as Bound, Jaeger, and Baker (1993) note, "the cure can be worse than the disease." To further substantiate an IV's relevance and adequacy, researchers need to test whether its inclusion in the first-stage regression (Equation 2) improves the model's R^2 value significantly, by referring to the F-statistic. As a rule of thumb, an F-statistic value of 10 or higher is generally viewed as indicative that the IV is adequate (e.g., Staiger and Stock 1997). Alternatively, Stock, Wright, and Yogo (2002) provide a table of critical F-statistic values for IV models based on the number of instruments used. Note that in situations with multiple endogenous variables and multiple IVs, researchers must apply a multivariate F-test (e.g., Angrist and Pischke 2009) to test for the IVs' adequacy. In addition, situations with multiple endogenous variables require researchers to consider at least the same number of IVs to ensure that the approach draws on an identified, or overidentified, model (Murray 2006b).

Second, according to the exclusion restriction for evaluating IVs when examining endogeneity, each IV should not be correlated with the error term in the original regression model (Murray 2006b). This requirement cannot be assessed statistically unless the researcher specifies more IVs than endogenous variables. In this case, the Sargan test and the Hansen J test enable researchers to test for significant correlations between the IVs and the error term (Bascle 2008; Wooldridge 2010). Rejecting the null hypothesis of no correlation, however, does not help to verify which of the IVs are endogenous (Papies, Ebbes, and van Heerde 2017). Although this test provides some empirical substantiation when the first-stage model is overidentified, researchers should rely primarily on theoretical arguments when assessing the IVs' adequacy (e.g., Murray 2006a).

Researchers have proposed the control function approach, a modification of the 2SLS method, which also leverages IVs to control for endogeneity. Specifically, the control function approach draws on the first-stage regression (Equation 2) to obtain the fitted residuals ($\hat{x}_1 - x_1$), which are then included in the original model to obtain correct(ed) estimates of the coefficients. Furthermore, interpreting the significance of the fitted residuals requires running the Hausman test (Papies, Ebbes, and van Heerde 2017). That is, if a coefficient is significant, researchers should revert to the endogeneity-corrected estimates.

The control function approach is particularly advantageous when regression models with discrete dependent variables are estimated (Petrin and Train 2010). However, such model types currently cannot be estimated in PLS-SEM because the standard application of the algorithm requires the dependent constructs to be continuous (Lohmöller 1989, Chapter 4). For linear models as typically estimated in PLS-SEM, the control function approach yields exactly the same results as the 2SLS approach (Papies, Ebbes, and van Heerde 2017).

Instrumental Variable-Free Approaches

When the IV approach is used, identification of conceptually fitting IVs is challenging (Rossi 2014), especially in light of the data constraints characterizing a typical research project. To address this problem, researchers have suggested IV-free approaches that do not require IVs to be specified. For example, Park and Gupta (2012) introduced the Gaussian copula approach, which controls for endogeneity by directly modeling the correlation between the endogenous variable and the error term by means of a copula. Another IV-free approach, proposed by Ebbes et al. (2005), is conceptually similar to the IV approach in that it splits an endogenous variable into exogenous and endogenous parts. However, this latent instrumental variable (LIV) approach does not draw on an observed variable but generates an unobserved (i.e., latent) IV. Ebbes et al. (2005) assume that the LIV is discrete and has at least two categories, separating the sample into L latent categories. Thus, with bivariate normally distributed error terms, the likelihood-based LIV approach belongs to the class of normal mixture models with L components (Papies, Ebbes, and van Heerde 2017) whose model parameters can be identified for $L > 1$ and nonnormality of the endogenous regressor (Ebbes et al. 2005).

The LIV approach can be used in a PLS-SEM context, but a fundamental challenge in its application is the selection of the number of categories, L . A priori, this number is

unknown, but it has a substantial effect on the results. Although Papies, Ebbes, and van Heerde (2017, p. 609) suggest that “the researcher fits at least $L = 2, 3, 4$ and 5 ,” there is no clear guidance regarding the desired stability of the estimates to vary L and thus regarding the final choice of this parameter. At the same time, whereas the statistical software R facilitates the Gaussian copula approach, current software support for the LIV approach is limited because models can include only one independent and one dependent variable (REndo version 1.3). Because these two issues limit the LIV approach’s applicability, we focus on the Gaussian copula approach, which enables correction for endogeneity in the absence of IVs.

Table 1 provides an overview of the advantages and disadvantages of the control variable approach, the IV approach, and two types of IV-free approaches (Gaussian copula and LIV) when dealing with endogeneity in PLS-SEM.

ADDRESSING ENDOGENEITY IN PLS-SEM

Procedure

PLS-SEM is a commonly used method to estimate path models with latent variables (Lohmöller 1989, Chapter 2; Wold 1982). Unlike factor-based SEM, which considers constructs as common factors that explain the covariation between the associated indicators (Jöreskog 1978), PLS-SEM is a composite-based approach to SEM that uses linear combinations of indicator variables to represent conceptual variables in a statistical model (Henseler et al. 2014; Sarstedt et al. 2016). PLS-SEM estimates the parameters of a set of equations in a structural equation model by combining an approach similar to principal components analysis with regression-based path analysis. Specifically, the PLS-SEM method uses partial regressions to estimate the path coefficients between the latent variables and their indicators in the measurement models, as well as between the latent variables in the structural model. Lohmöller (1989, Chapter 2) provides an introduction to the PLS-SEM algorithm. Approaches for dealing with endogeneity in regression models can therefore be readily transferred to PLS-SEM. However, in contrast to regression analysis, in which the approaches use the (observed) indicator variables as input, in PLS-SEM the implementation of these variables draws on the composite scores obtained after the algorithm’s convergence (Lohmöller 1989, Chapter 2).² Figure 1 illustrates our systematic procedure for dealing

with endogeneity in PLS-SEM. In the following sections, we discuss each step in greater detail.

Stage 1 of the procedure requires checking whether testing for endogeneity is necessary. Specifically, the meaningfulness of considering endogeneity in PLS-SEM depends on whether researchers use the method for predictive or explanatory modeling purposes. In predictive modeling, the purpose of the analysis is to predict new or future observations (Shmueli 2010). Here, the main concern is predicting the case values of the dependent variables, rather than (causally) interpreting the structural paths. Controlling for endogeneity is not useful in this case, as it may reduce the model’s predictive power (Ebbes et al. 2005). Conversely, in explanatory modeling, which involves “the use of statistical models for testing causal explanations” (Shmueli 2010, p. 290), controlling for endogeneity is crucial to adequately test hypotheses (Papies, Ebbes, and van Heerde 2017). In practice, however, researchers using PLS-SEM generally balance prediction and explanation in that they expect their model to have high predictive accuracy while being grounded in well-developed causal explanations (Sarstedt, Ringle, and Hair 2017a). Gregor (2006, p. 626) refers to this interplay as explanation and prediction theory, noting that this approach “implies both understanding of underlying causes and prediction, as well as description of theoretical constructs and the relationships among them.” Such a setting may require the consideration of endogeneity, depending on whether explanation (i.e., testing a theory) or prediction (e.g., deriving managerial recommendations) is the prevailing intent of a certain research project.

Stage 2 of the procedure (Figure 1) uses existing theory and prior research findings to identify potential endogeneity issues a priori. If prior research adequately theorized, detected, and treated endogeneity issues in a specific research context, researchers should apply this knowledge when proposing a PLS path model and collecting data. This method typically involves including control variables in the model, or gathering data on IVs for an ex post assessment, using the IV approach (e.g., Ebbes, Papies, and Van Heerde 2017; Reeb, Sakakibara, and Mahmood 2012). Thus, if prior research suggests that endogeneity is an issue, or if the results cast doubt on the assertion that all sources of endogeneity have been detected, researchers should test whether endogeneity substantially affects the model’s estimates.

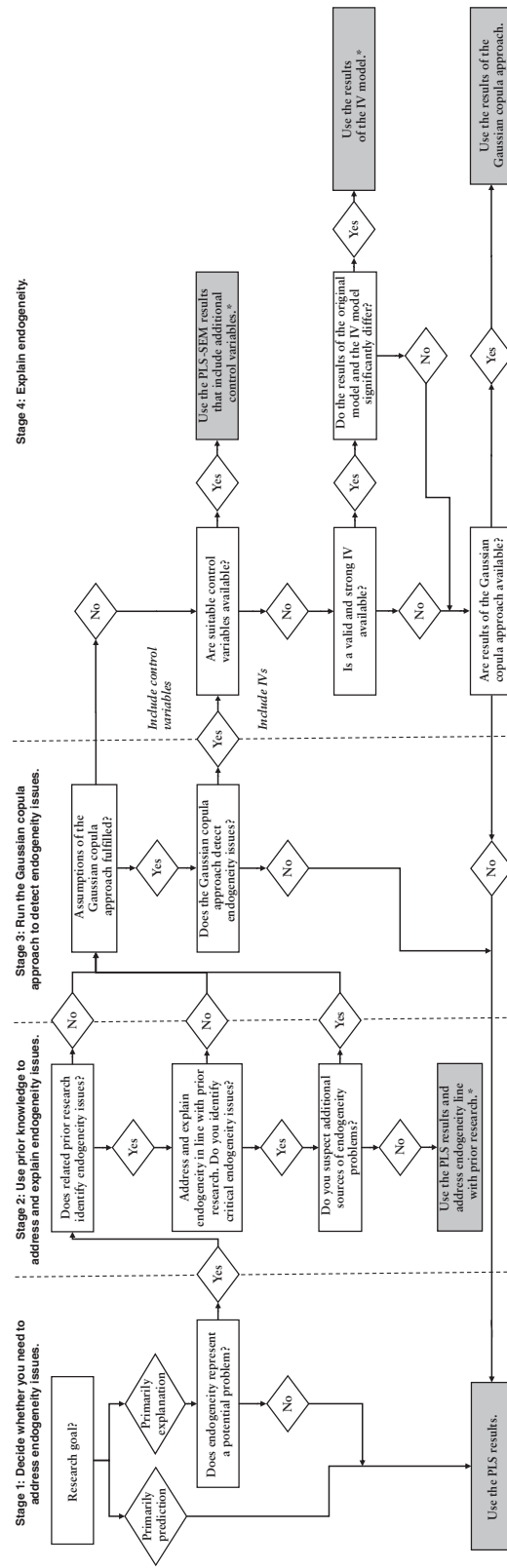
To do so, in Stage 3 of the procedure (Figure 1), researchers apply the Gaussian copula approach. If the results do not indicate a problem, researchers can use the original model’s

²This implementation is similar to other methodological extensions, such as finite mixture PLS (Hair et al. 2018), which uses composite scores as input for mixture regressions.

Table 1. Comparison of Approaches to Deal with Endogeneity

Criteria	Methods			
	Control Variable Approach	Instrumental Variable Approach	Instrumental Variable-Free Approaches	
			Gaussian Copula	Latent Instrumental Variable
Number of variables	Data on additional variables must be collected	Instrumental variables have to be identified, and data must be collected	No additional variables needed	No additional variables needed
Distribution of variables	No assumptions required	No assumptions required	Endogenous variables have to be nonnormally distributed	Endogenous variables have to be nonnormally distributed
Nature of dependent variable	Discrete or continuous	Continuous	Discrete or continuous	Continuous
Statistical tests	Not necessary	Test for significance and relevance	Test for significance	Test for significance
Acceptance in scientific community	Widely accepted and commonly used	Widely accepted and commonly used	Relatively new and therefore rarely used	Relatively new and therefore rarely used
Implementation in software	No additional implementation necessary	Supported by, for example, SPSS, Stata, and R software packages	Supported by the REndo (Gui, Meierer, and Algesheimer 2017) package	Supported by the REndo (Gui, Meierer, and Algesheimer 2017) package with one dependent variable and one independent variable

Figure 1. Procedure to Address Endogeneity in PLS-SEM



* Consider continuing at Stage 3 and using the Gaussian copula approach to assess the extent to which the endogeneity problem has been successfully addressed by the model.

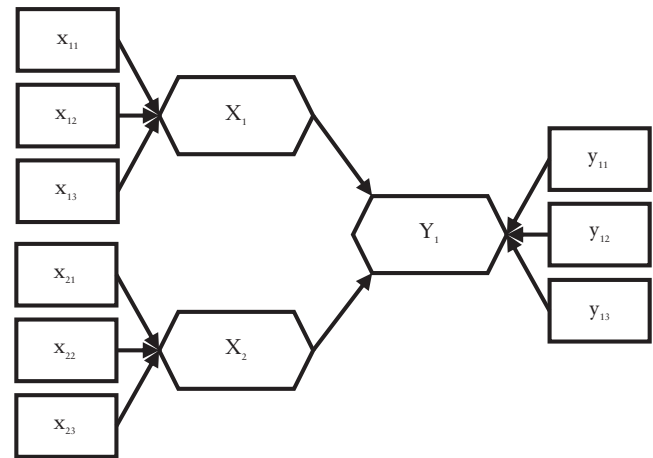
estimates. Otherwise, the sources of endogeneity must be further explored and controlled for in the next step of the procedure. Because endogeneity problems in international marketing often arise from omitted variables (Jean et al. 2016), Stage 4 involves conceptually identifying the missing information and adding control variables to the PLS path model (Papies, Ebbes, and van Heerde 2017). The inclusion of suitable control variables deals with, or at least reduces, the impact of endogeneity on the model estimates (Ebbes, Papies, and van Heerde 2017). In situations in which no suitable control variables are available, researchers can draw on the IV approach. If IVs are unavailable, or if researchers cannot identify at least one suitable IV for each endogenous variable in a partial regression model, they can use the Gaussian copula results, if available, and report that they cannot identify omitted variables that adequately explain the endogeneity problem. If Gaussian copula results are not available, because they do not meet the method's requirements, researchers should interpret the original PLS path model estimates and specify the reason(s) for not addressing the potential endogeneity issues as a limitation (Figure 1).

The Gaussian Copula Approach in PLS-SEM

To illustrate the Gaussian copula approach in PLS-SEM, consider a simple path model with one dependent latent variable, Y_1 , explained by two independent latent variables, X_1 and X_2 . Each of the latent variables is measured using three indicators. Figure 2 shows this simple path model. The hexagons represent composites formed by linear combinations of the indicator variables, as indicated by the arrows linking the indicators and the composites. These arrows do not necessarily correspond to the measurement model specification, which can be reflective or formative (Sarstedt et al. 2016). However, we use this illustration because PLS-SEM forms composites from the indicator variables to represent the conceptual variables in the model estimation.

Model estimation in PLS-SEM draws on a three-stage approach that belongs to the family of (alternating) least squares algorithms (Mateos-Aparicio 2011). Stage 1 of the PLS-SEM algorithm iteratively determines indicator weights, composite scores, and path coefficients. The algorithm converges when the indicator weights change only marginally from one iteration to the next. Stages 2 and 3 use the final composite scores from Stage 1 as input for a series of ordinary least squares regressions. These regressions produce the final indicator loadings, indicator weights, and path coefficients, as well as related elements, such as the indirect and total effects, and R^2 values (e.g., Sarstedt,

Figure 2. Simple PLS Path Model



Ringle, and Hair 2017a). With reference to our example, the final stage of the PLS-SEM algorithm entails regressing Y on X_1 and X_2 to estimate the structural model path coefficients β_1 and β_2 :

$$(4) \quad Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

Let us assume that the independent variable X_1 is thought to be correlated with the error term (ϵ) of the dependent variable Y and may therefore induce an endogeneity problem in the estimation of Equation 4.

The Gaussian copula approach requires the composite scores of the endogenous construct X_1 to be nonnormally distributed. If this requirement is met, the Gaussian copula of X_1 can be calculated as follows (Papies, Ebbes, and van Heerde 2017):

$$(5) \quad c^* = \Phi^{-1}[H(X_1)],$$

where $H(x)$ is the empirical cumulative density function and Φ^{-1} is the inverse normal cumulative density function. Next, the copula is included as an additional independent variable that controls for the correlation between the error term and the endogenous independent construct in the regression model:

$$(6) \quad y = \beta_0 + \beta_1' x_1 + \beta_2' x_2 + \beta_3 c_1^* + \epsilon'.$$

To attend to the endogeneity problem, estimating Equation 6 yields the new path coefficients β_1' and β_2' .

To determine whether endogeneity is at a critical level, researchers need to assess the significance of the copula coefficient (β_3) using bootstrapping (Hausman 1978; Park and Gupta 2012). A significant copula coefficient indicates

a critical level of endogeneity. If the copula coefficient is significant, researchers can assume that an endogeneity issue is present, even though it is treated by including the copula in the regression model (Equation 6). If the copula coefficient is not significant, no critical endogeneity issue affects the regression results.

Although the standard Gaussian copula approach assumes a single endogenous variable, Park and Gupta (2012) discuss the case of modeling multiple endogenous variables simultaneously. Doing so requires creating the copula for each individual endogenous variable and adding all copulas to the original model. The procedure separately computes all the Gaussian copulas of the specified endogenous variables (Park and Gupta 2012).

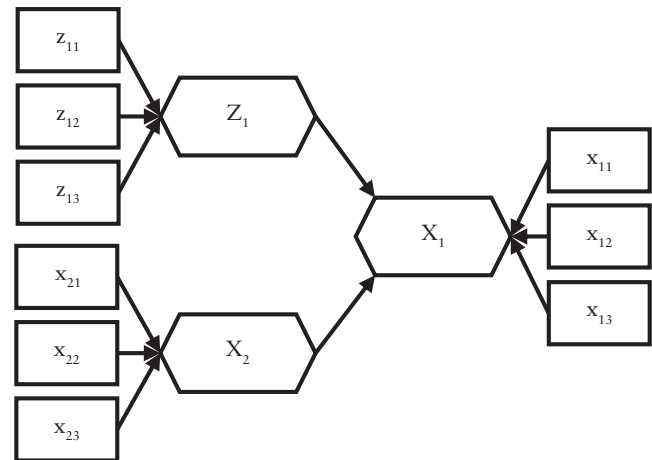
Control Variables in PLS-SEM

If the Gaussian copula approach indicates a significant degree of endogeneity, researchers should use control variables to identify its sources. The introduction of control variables should always be grounded in strong theory (Bernerth and Aguinis 2016). That is, researchers should be explicit (rather than implicit) regarding the role of control variables in the analysis and potentially even match the hypotheses precisely to their choice (Spector and Brannick 2010). Typical examples of control variables used in international marketing models include a company's location, age, size, export experience, years of global experience, percentage of products sold overseas, number of countries entered, and cultural distance (e.g., Cui, Walsh, and Zou 2014; Shu, Jin, and Zhou 2017).

The use of control variables in PLS-SEM is analogous to their use in regression models. More precisely, control variables are included as single-item constructs in a PLS path model. Each control variable is then linked to the dependent latent variable whose predictor constructs have been identified as (or are assumed to be) possibly exhibiting endogeneity. Note that criticism of the use of single-item constructs (Diamantopoulos et al. 2012) does not apply to control variables, because they usually represent observable characteristics. However, if the control variables represent unobservable concepts (e.g., cultural dimensions), they need to be measured by means of multiple items. Finally, the interpretation of the control variables' effect is also analogous to that in regression models and depends on each control variable's measurement scale, which can be categorical or metric in PLS-SEM (Hair et al. 2017).

After the control variables are specified, researchers need to use bootstrapping to assess these variables' significance.

Figure 3. First Stage of the IV Approach



Similar to the analysis of interaction effects in PLS-SEM (Hair et al. 2017; Chapter 7), the assessment of significant control variables should include the use of the f^2 effect size to assess their relevance. According to Chin (1998), f^2 values of .02, .15, and .35 suggest small, moderate, and substantial effects.

The Instrumental Variable Approach in PLS-SEM

Implementing the IV approach in the context of PLS-SEM requires adjusting the final stage of the PLS-SEM algorithm. In its original form, this stage uses the composite scores obtained after convergence as input for a series of ordinary least squares regressions that estimate the structural model path coefficients. Instead of using ordinary least squares regressions, however, this stage now uses the 2SLS method (Benitez, Henseler, and Roldán 2016). When implementing the IV approach in PLS-SEM, researchers need to ensure that they include at least one IV per endogenous variable.

Analogous to the process of regression analysis, the IV approach's implementation requires researchers to find IVs that are highly correlated with the independent variables (instrument relevance) but are uncorrelated with the dependent variable's error term (exclusion restriction). In international marketing, previous regression-based models used IVs such as organization quality, brand quality, environmental causes, brand strength (Swoboda and Hirschmann 2017), exposure to a multinational company's marketing (Swoboda and Hirschmann 2016), and export memory (Souchon et al. 2016). However, the identification of suitable IVs is highly context-specific and is the most challenging step in the analysis (e.g., Larcker and Rusticus

2010). Angrist and Pischke (2009, p. 17) argue that finding good IVs requires “a combination of institutional knowledge and ideas about processes determining the variable of interest,” especially because no true test of their quality is available (Rossi 2014).

To illustrate the implementation of the IV approach in PLS-SEM, consider the simple model in Figure 2 with the construct X_1 possibly exhibiting endogeneity. We now introduce an IV (i.e., Z_1), measured by three indicators, that correlates strongly with the construct X_1 but not with the error term of Y_1 in the original model. Thereafter, we create and estimate the model shown in Figure 3.

Next, a 2SLS analysis uses the composite scores of X_1 and X_2 , obtained from the first model (Figure 2), and the composite scores of Z_1 , derived from the second model (Figure 3). Specifically, we regress X_1 on Z_1 and X_2 to obtain $\hat{X}_1$:

$$(7) \quad \hat{X}_1 = \gamma_0 + \gamma_1 Z_1 + \gamma_2 X_2.$$

We first assess the IV's strength by comparing the R^2 values of the first-stage regression (Equation 7) with and without the IV. The results of an F-test indicate whether the R^2 values of the regression model with and without the IVs differ significantly. If significant differences are found, we continue with the analysis by theoretically substantiating that Z_1 is not correlated with the error term of Y_1 (e.g., Semadeni, Withers, and Certo 2014; Staiger and Stock 1997). In the case of an overidentified model, statistical assessments by means of the Sargan test and the Hansen J test may support this analysis.

Finally, we reestimate the original model by regressing Y on X_2 and $\hat{X}_1$ from Equation 7 to obtain the corrected path coefficient estimates β'_1 and β'_2 :

$$(8) \quad Y = \beta'_0 + \beta'_1 \hat{X}_1 + \beta'_2 X_2 + \epsilon'.$$

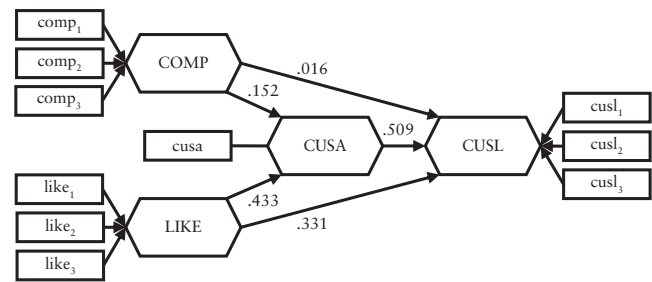
The Hausman test (Hausman 1978) enables us to test whether β'_1 (Equation 8) differs significantly from β_1 (Equation 5). If so, researchers should revert to the IV approach's coefficients (Equation 8). If the difference is not significant, researchers should continue using the results of the partial regression model's original PLS-SEM estimation.

EMPIRICAL EXAMPLE TO ADDRESS ENDOGENEITY IN PLS-SEM

Model and Data

To illustrate our procedure for identifying and dealing with endogeneity (Figure 1), we draw on a corporate reputation

Figure 4. The Simple Corporate Reputation Model



model originally presented by Eberl (2010). The goal of this model is to explain the effects of competence (COMP) and likability (LIKE), which represent corporate reputation's two dimensions (Sarstedt, Wilczynski, and Melewar 2013; Schwaiger 2004), on customer satisfaction (CUSA) and ultimately on customer loyalty (CUSL). This model has been used to illustrate aspects of the PLS-SEM method (e.g., Hair et al. 2017; Hair et al. 2018) and has been frequently applied in an international marketing context.³ For example, Zhang and Schwaiger (2012) use this model to study the differentiating effects of corporate social responsibility and other antecedent dimensions on corporate reputation in three Western countries (Germany, the United States, and the United Kingdom) and in China. Similarly, Radomir and Alan (2018) apply this model to contrast the effects of service quality and relationship investment on corporate reputation assessments of Romanian and U.K. consumers. Furthermore, Walsh, Beatty, and Shiu (2009) developed and validated a variant of this model using cross-country data; their model features prominently in the international marketing literature. For example, several researchers have used this model to analyze the moderating effect of institutional country differences (Swoboda, Puchert, and Morschett 2016) and cultural dimensions (Swoboda and Hirschmann 2017) on the relationship between corporate reputation and customer loyalty.

Figure 4 shows the original model, which includes two partial regression models: (1) CUSL is regressed on COMP, CUSA, and LIKE, while (2) CUSA is regressed on COMP and LIKE. The following illustrations focus on the first, more complex, regression model. The measurement models of COMP, LIKE, and CUSL draw on three reflective items each, whereas CUSA is measured with a single item. The model estimation draws on data from two

³The model and data can be downloaded from <https://www.pls-sem.com/>.

major German mobile communications network providers and two smaller competitors. A total of 344 respondents rated each item on a seven-point Likert scale. Observations with missing values were deleted, leaving a total sample size of 336. We used the statistical software R (R Core Team 2018), Stata (StataCorp 2015), and SmartPLS 3 (Ringle, Wende, and Becker 2015) for the computations. For the PLS-SEM estimations, we use Mode A for all composites (Sarstedt, Ringle, and Hair 2017a). The results of the bootstrap routine with 5,000 samples, with the no-sign-changes option applied, enable us to test for the coefficients' significance. We find that the measurement models meet all the relevant evaluation criteria. In the structural model assessment, all the path coefficient estimates, except COMP to CUSL, are significant, and the model's R^2 values are satisfactory (for this application, Hair et al. [2017] provide further details of the PLS-SEM results and their assessment).

Most corporate reputation model analyses take a predictive perspective (e.g., Schwaiger 2004). However, in line with Eberl (2010), who elaborates on the causal effects of corporate reputation's two components (i.e., COMP and LIKE) on CUSA and CUSL, we consider our analysis primarily explanatory in nature. Thus, following Stage 1 of the procedure (Figure 1), we continue by applying the Gaussian copula approach.

Application of the Gaussian Copula Approach in PLS-SEM

Before initiating the Gaussian copula approach, to meet its assumptions, we first verify whether the variables that potentially exhibit endogeneity are nonnormally distributed (Figure 1). We do so by running the Kolmogorov–Smirnov test with Lilliefors correction (Sarstedt and Mooi 2014) on the standardized composite scores of COMP, LIKE, and CUSA, which the PLS path model estimation provides. The results show that none of the constructs have normally distributed scores; thus, we can consider them endogenous in the Gaussian copula analysis.

Next, we create three regression models in which we consider the independent constructs COMP (model 1), LIKE (model 2), and CUSA (model 3) as possibly exhibiting endogeneity. In addition, to simultaneously account for multiple endogenous variables when applying the Gaussian copula approach, we create four regression models that include all possible combinations of multiple endogenous variables in this PLS path model example: LIKE and COMP (model 4), LIKE and CUSA (model 5), COMP and CUSA (model 6), and LIKE, COMP, and CUSA (model 7).

We use the constructs' standardized composite scores to compute the Gaussian copula of the partial regressions in the structural model. To cross-validate the results, we used two different statistical software packages, which returned the same results: Stata 14 (StataCorp 2015) and the Rendo package of the R program (Gui, Meierer, and Algesheimer 2017). To implement Stata, we used specific commands (1) to compute the Gaussian cumulative distribution function, (2) to create the copula as the inverse of the Gaussian normal cumulative distribution function, and (3) to calculate the model that includes the Gaussian copula. Because the Rendo package does not currently support the computation of bootstrap standard errors, we used our own R code and the boot package (Canty and Ripley 2017) to run this part of the analysis.⁴

The results in Table 2 show that only one Gaussian copula (i.e., c_{CUSA}) is significant ($p < .1$) when we consider one endogenous variable; this finding points to a potential endogeneity issue. Including the significant Gaussian copula in the model changes the effect of CUSA on CUSL by .073 units (from .509 to .582), which points to a potential endogeneity problem for CUSA (model 3). Similarly, c_{CUSA} is also significant in the CUSA models in combination with LIKE (model 5, $p < .1$), COMP (model 6, $p < .05$), and LIKE and COMP (model 7, $p < .1$). This finding confirms the possibility of CUSA being endogenous.⁵

Application of the Control Variable Approach in PLS-SEM

After completing Stage 3 of the procedure for identifying and treating endogeneity (Figure 1), we next include in the model control variables that, from a theoretical perspective, are likely to influence customers' loyalty (Eberl 2010). The first control variable (serviceprovider) distinguishes between large, globally operating mobile phone service providers (serviceprovider = 1; $n_{(1)} = 251$) and smaller, locally operating competitors (serviceprovider = 2; $n_{(2)} = 85$). In addition, we include a second control variable (servicetype), which is relevant from a service marketing perspective and distinguishes between customers with a contract and those with a prepaid plan.

Analysis of the impact of the two control variables, both separately and jointly, shows that neither has a significant

⁴The R code can be downloaded at <https://www.pls-sem.com/>.

⁵The parallel findings from the application of the Gaussian copula approach to single and multiple endogenous variables is not surprising, since the technique computes the copulas separately when considering multiple endogenous variables in the PLS path model.

Table 2. Results of the Gaussian Copula Approach

Variable	Original Model		Gaussian Copula Model 1 (Endogenous Variable: COMP)		Gaussian Copula Model 2 (Endogenous Variable: LIKE)		Gaussian Copula Model 3 (Endogenous Variable: CUSA)		Gaussian Copula Model 4 (Endogenous Variables: LIKE, COMP)		Gaussian Copula Model 5 (Endogenous Variables: LIKE, CUSA)		Gaussian Copula Model 6 (Endogenous Variables: COMP, CUSA)		Gaussian Copula Model 7 (Endogenous Variables: LIKE, COMP, CUSA)	
	Value	p-Value	Value	p-Value	Value	p-Value	Value	p-Value	Value	p-Value	Value	p-Value	Value	p-Value	Value	p-Value
COMP	.016	.746	.014	.848	.017	.763	.021	.707	-.006	.939	.021	.705	-.027	.717	-.037	.653
LIKE	.331	<.01	.331	<.01	.370	<.01	.331	<.01	.381	<.01	.341	<.01	.333	<.01	.362	<.01
CUSA	.509	<.01	.509	<.01	.511	<.01	.582	<.01	.509	<.01	.580	<.01	.592	<.01	.589	<.01
c _{COMP}			.002	.974					.019	.737			.039	.432	.047	.405
c _{LIKE}					-.033	.245			-.041	.283	-.008	.790			-.024	.509
c _{CUSA}							-.041	.063			-.039	.084	-.049	.045	-.047	.061

Table 3. Results of the IV Approach

Endogenous Variable	Coefficients			R ² Values		Weak ID Test		Hausman Test
	COMP	LIKE	CUSA	First Stage Without IV	First Stage with IV	F-Value	Significant?	p-Value
LIKE	-.075	.506	.454	.496	.560	49.478	Yes	.216
COMP	.120	.278	.490	.433	.560	97.647	Yes	.216
CUSA	-.087	.081	1.113	.295	.300	2.575	No	.216

influence on the CUSL construct. Specifically, the servicetype variable has a $-.003$ effect (p -value = .917; $f^2 = .000$) and the service provider variable has a $-.038$ effect (p -value = .319; $f^2 = .003$) on CUSL when included separately. Including both control variables simultaneously changes the results only marginally. Because none of the control variables identify a source of endogeneity, we continue with the procedure (Figure 1) by running the IV approach on the data.

Application of the IV Approach in PLS-SEM

The application of the IV approach in the context of our model requires selection of an IV that is correlated with COMP, CUSA, and LIKE. At the same time, the IV needs to be uncorrelated with the omitted determinants of customer loyalty and thus be uncorrelated with CUSL's error term. Our implementation of the IV approach uses customers' perceived switching costs as an IV. Switching costs comprise the time, money, and effort that consumers expect to expend when changing their service provider. In the mobile phone market, switching costs come primarily in the form of transfer costs (Klemperer 1987) incurred when a consumer ends a relationship with one provider and starts a new one with another. Specifically, the implementation and configuration of the new relationship require the customer to invest time, effort, and other assets, while ending an old service relationship may also require procedural tasks (Ray, Kim, and Morris 2012). Thus, in terms of the instrument relevance criterion, it is reasonable to assume that the perceived switching costs correlate with the customers' satisfaction and reputation assessments, because significant transfer costs limit their freedom to configure the business relationship. Given the manifold factors that influence loyalty, arguing that perceived switching costs are uncorrelated with omitted variables that affect customer loyalty (exclusion restriction) is more difficult. However, research has identified customers' variety-seeking tendency—the “tendency of individuals to seek diversity in their choices of services or goods” (Kahn 1995, p. 139)—as a major

determinant of customer loyalty (e.g., Berné, Múgica, and Yagüe 2001). Because this tendency is a psychological trait that is independent of individuals' preference structures and is stable over time (Menon and Kahn 1995), variety seeking is probably unrelated to customers' perception of switching costs. Network quality is another key determinant of customer loyalty that is relevant in the context of our study (e.g., in terms of coverage, stability, and voice quality; Lim, Widdows, and Park 2006). Network quality can differ substantially, especially in rural areas, and cannot be fully observed a priori. Poor network coverage will have a strong negative effect on customer loyalty, independent of the customers' perception of the switching costs.⁶

We use an extended version of Jones, Mothersbaugh, and Beatty's (2000) scale and measure the switching costs by using four reflective indicators (switch₁ to switch₄), each measured on a five-point Likert scale. For our analysis, we extract all (standardized) composite scores from the SmartPLS 3 software. We subsequently run the IV approach by using Stata's ivregress method (StataCorp 2017), as well as the statistical software R with its AER package (Kleiber and Zeileis 2017) to validate the results. Both programs provide the same results. Table 3 shows the results of the analysis for the IVs.

To assess whether the IV is strong (Figure 1), we use the weak ID test's F-value, which compares the R² values of the first-stage regression with and without the IV. We consider three models, each of which considers another predictor construct of CUSL as likely to exhibit endogeneity (i.e., LIKE, COMP, and CUSA). The results in Table 3 show that including the IV significantly increases the first-stage

⁶While these arguments support the use of perceived switching costs, we note that this is not an ideal IV, as one could expect this variable to correlate with other service perceptions (e.g., perceived service quality) that are relevant for customer loyalty and which the two reputation dimensions capture only indirectly (e.g., Raithel and Schwaiger 2015).

R^2 value when COMP and LIKE are considered as endogenous, as evidenced by the weak ID test values well above 10. Conversely, the model's R^2 value increases only marginally when CUSA is considered likely to exhibit endogeneity. Thus, in the following step, we use the Hausman test to assess whether the coefficients of the IV approach differ significantly from the original coefficients. The results in Table 3 show that, in our analysis, the original and IV coefficients do not differ significantly. When LIKE and COMP are considered endogenous, the Hausman test specifically returns p -values of .216.⁷

Including switching costs as an IV does not explain the endogeneity problem's origin. More specifically, the IV correction of CUSA increases the corresponding point estimate to greater than one, further indicating that the switching costs construct is not a suitable IV.

Following the procedure in Figure 1, we examine the results of the Gaussian copula approach. Specifically, because the inclusion of CUSA's significant Gaussian copula in the model changes the relationship of CUSA and CUSL from .509 to .582, researchers should consider CUSA as endogenous (Table 2). This finding is in line with prior research that has theoretically and empirically substantiated endogeneity problems in studies of customer satisfaction, especially its relationship to customer loyalty (Ivanov, Joseph, and Wintoki 2013). Nevertheless, this analysis illustrates that, in the context of our empirical example, considering endogeneity does not, from a theory-testing perspective, change any conclusions drawn from the model estimation.

To summarize, by carrying out the procedure shown in Figure 1, we identified a potential endogeneity issue based on the Gaussian copula results. However, we could not explain the problem in light of the available control and instrumental variables. In this situation, the researcher should report the solution found with the Gaussian copula approach, acknowledging that the findings do not reveal the origins of the endogeneity.

SUMMARY AND CONCLUSIONS

Although addressing endogeneity has become an integral component of regression-based analyses in international marketing (Jean et al. 2016), users of the PLS-SEM method

⁷We also ran the control function approach on the model using switching costs as the IV. We used the statistical software R for the calculation and the boot package to bootstrap the standard errors (Canty and Ripley 2017). As expected, the results parallel those of the IV approach.

have overlooked this issue. A potential reason for this disregard is the predictive nature of PLS-SEM, which, if taken into full consideration when the method is used, renders endogeneity irrelevant (Ebbes, Papies, and van Heerde 2011). However, researchers conventionally use PLS-SEM to test formal hypotheses, which corresponds to an explanatory perspective, while simultaneously deriving managerial implications, which follows a predictive perspective. Endogeneity becomes an important concern, therefore, depending on whether researchers emphasize explanation or prediction in their analyses.

Similar to scholars who have applied PLS-SEM in international marketing and other disciplines, methodological researchers have failed to consider endogeneity in PLS-SEM, with some authors suggesting that endogeneity cannot be accounted for in a PLS-SEM framework (e.g., Antonakis et al. 2010; Rönkkö and Evermann 2013). Such notions are astonishing because PLS-SEM is a regression-based technique, which makes common approaches for identifying and treating endogeneity accessible in this specific methodological context. Benitez, Henseler, and Roldán (2016) took a first step in this direction by combining the standard PLS-SEM algorithm with the 2SLS method; however, they did not consider actual IVs that allow the independent variables' variance to be decomposed into endogenous and exogenous parts. Broadening the scope of their research, this study introduces a systematic procedure for identifying and treating endogeneity in PLS-SEM that takes into account the most recent statistical developments in the literature on endogeneity. Specifically, implementing the Gaussian copula approach addresses the significant concerns associated with identifying strong and valid IVs. Moreover, combining this novel approach with the inclusion of control variables and the classic IV approach enables a comprehensive assessment of endogeneity when PLS-SEM is applied for explanatory purposes.

Our application of the procedure to empirical data reveals endogeneity in Eberl's (2010) corporate reputation model, which has been frequently used in different variations in international marketing and international business research (e.g., Swoboda, Puchert, and Morschett 2016; Zhang and Schwaiger 2012). Our analysis suggests that the CUSA construct is subject to endogeneity, but control variables or IVs were not sufficient to handle this problem. Correcting for endogeneity with the use of the Gaussian copula approach leads to a pronounced change in the model estimates, while leaving the general conclusions intact.

We hope that this article will be the starting point for future PLS-SEM research on endogeneity. Researchers may

APPENDIX. International Marketing Studies That Apply PLS-SEM

Year	Journal		
	<i>International Marketing Review</i>	<i>Journal of International Business Studies</i>	<i>Journal of International Marketing</i>
2017	Gomes et al. (2017); Moon and Oh (2017); Pinho and Thompson (2017); Rahman et al. (2017)		Ashraf et al. (2017); Obadia, Vida, and Pla-Barber (2017)
2016	Andéhn and L'Espoir Decosta (2016); Henseler et al. (2016b); Jean et al. (2016); Richter et al. (2016); Schlegelmilch et al. (2016)	Lew et al. (2016)	
2015	Rippé et al. (2015); Sinkovics et al. (2015)	Khan et al. (2015); Obadia et al. (2015)	Griffith and Yanhui (2015); Morgeson et al. (2015); Sheng et al. (2015)
2014	Freeman and Styles (2014); Griffith et al. (2014)		Ashraf et al. (2014); Diamantopoulos et al. (2014)
2013	Sinkovics et al. (2013)	Schotter and Beamish (2013)	Magnusson et al. (2013); Obadia (2013); Swoboda and Elsner (2013)
2012		Lam et al. (2012)	Hilmersson and Jansson (2012); Johnston et al. (2012)
2011			Hortinha et al. (2011); Nijssen and Douglas (2011); Sichtmann et al. (2011)
2010	Duque and Lado (2010)	Shi et al. (2010)	Navarro et al. (2010); Sichtmann and von Selasinsky (2010)
2009		Fey et al. (2009)	Lages et al. (2009); Nijssen and Herk (2009)
2008			Brettel et al. (2008); Nijssen and Douglas (2008)
Number of articles (2008–2017)	15	7	21

endeavor to introduce bootstrapping-based approaches to test the strength of an IV, its validity, and its relevance in PLS-SEM. Further, the weak ID test and the Hausman test are parametric tests and therefore violate the nonparametric nature of PLS-SEM, thus requiring other tests to be identified. Moreover, further research should advance the understanding and applicability of IV-free approaches, such as Gaussian copulas or Ebbes et al.'s (2005) LIV method. For example, simulation study results could provide researchers with thresholds for deciding whether a significant Gaussian copula entails a negligible endogeneity problem or a substantial one. Future research should also identify ways to parameterize the LIV approach, which we did not consider in our study because of its strong dependence on the number of latent instrument categories (Papies, Ebbes, and Van Heerde 2017) and lack of software support. Such research would pave the way for a benchmarking study of the Gaussian copula and LIV approaches in the context of PLS-SEM. In this regard, a Monte Carlo simulation study, in which prespecified levels of endogeneity bias the parameter estimates, would enable comparison of the IV and IV-free approaches' efficacy in identifying and treating endogeneity.

Recent PLS-SEM developments, such as the consistent PLS-SEM approach (Dijkstra and Henseler 2015), which allows for mimicking factor-based SEM results, and the introduction of model fit criteria into PLS-SEM (e.g., Henseler et al. 2014), may help researchers achieve specific explanatory research goals. The next logical step is to extend PLS-SEM's capabilities to address endogeneity issues in empirical international marketing research when the consistent PLS-SEM approach is used to estimate models on the grounds of common factor model data (Rigdon, Sarstedt, and Ringle 2017; Sarstedt et al. 2016). This advancement should be straightforward, in terms of our research findings, because the consistent PLS-SEM method uses the composite scores obtained from a standard PLS-SEM-based estimation as input.

Future research should also explore other approaches to detect endogeneity problems in PLS-SEM. Research has brought forward a number of latent-class techniques, such as finite mixture PLS (Hair et al. 2018), PLS prediction-oriented segmentation (Becker et al. 2013), and PLS iterative reweighted regression segmentation (Schlittgen et al. 2016), to uncover unobserved heterogeneity. Sarstedt, Ringle, and Hair (2017b) recently introduced a systematic procedure for identifying and treating unobserved heterogeneity in PLS path models. Combining several latent-class techniques, the procedure enables researchers to check whether unobserved heterogeneity has a critical impact on the results, decide on the number of latent segments to retain

from the data, and recover the segment-specific model estimates. However, researchers need to assess the interplay between the various sources of heterogeneity on the one hand and endogeneity on the other and develop guidelines to assess whether endogeneity is at a noncritical level in the segment-specific solutions.

Consideration of the fuzzy-set qualitative comparative analysis of regression models (Woodside 2013) would be particularly interesting. Joint consideration of these techniques may enable researchers to better assess and control for endogeneity embedded in unobserved heterogeneity (Ivanov, Joseph, and Wintoki 2013; Schlittgen et al. 2016).

Moreover, in this study we did not address how to deal with endogeneity issues in PLS path models that include interaction terms, which are used, for example, in PLS-SEM-based moderator analyses (Hair et al. 2017; Henseler and Chin 2010). Because international marketing researchers frequently apply such analyses (e.g., Jin, Zhou, and Wang 2016; Leonidou et al. 2017), future studies should use insights from research on regression analysis (e.g., by using special 2SLS analyses to account for interaction terms and IVs or the control function approach; see, for example, Sande and Ghosh 2018) to furnish recommendations on how to adequately address this issue in PLS-SEM.

Finally, in its standard form PLS-SEM does not support the estimation of longitudinal or panel data without limitations (Roemer 2016). After the PLS-SEM method advances in these directions, assessing the efficacy of other approaches for handling endogeneity, such as the lagged variable approach (Jean et al. 2016), the cross-lagged correlation procedure (Tyagi and Wotruba 1993), the Arellano–Bond (1991) estimator (Kumar and Pansari 2016), and the Granger causality test (Filipescu et al. 2013), would be particularly interesting. Extending the PLS-SEM algorithm to accommodate binary dependent variables would provide an opportunity to consider endogeneity methods specific to such model settings, such as the control function approach and its extensions (e.g., Garen 1984; Wooldridge 2015). These advances would be particularly beneficial in international marketing research, which frequently relies on longitudinal and panel data (e.g., Kumar and Pansari 2016) as well as binary outcome variables (e.g., Obadia, Vida, and Pla-Barber 2017; Westjohn, Roschk, and Magnusson 2017).

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